

Triple Peaks

Cordillera Oriental n' itsinda ry' imisozi iherereye muri Andes inyura muri Bolivia. Igizwe n' urutonde rw' udusongero (peaks) tw' imisozi N , itondekanyijwe guhera kuri 0 kugeza kuri $N - 1$. **Uburebure** bw' umusozi i ($0 \leq i < N$) ni $H[i]$, ni umubare (integer) uri hagati ya 1 na $N - 1$, inclusive.

Kuri buri peaks ebyiri i na j aho $0 \leq i < j < N$, **distance** iri hagati yazo ni $d(i, j) = j - i$.

Ukurikije abakurambere ba Inca legends, peaks eshatu ziba **mythical** iyo zifite ibi bikurikira: uburebure bwa peaks eshatu **buhwana** na distances ziri hagati ya buri misozi ibiri, **wiregagije uko distances zitondekanyije**.

Muri make, itsinda rya indices eshatu (i, j, k) ni mythical iyo

- $0 \leq i < j < k < N$, kandi
- uburebure $(H[i], H[j], H[k])$ buhwana na pairwise distances $(d(i, j), d(i, k), d(j, k))$ utitaye k' uko zitondekanyije. Urugero, indices 0, 1, 2 mu gihe pairwise distances ari (1, 2, 1), uburebure bushoboka ni $(H[0], H[1], H[2]) = (1, 1, 2)$, $(H[0], H[1], H[2]) = (1, 2, 1)$, na $(H[0], H[1], H[2]) = (2, 1, 1)$, gusa uburebure $(H[0], H[1], H[2]) = (1, 2, 2)$ bwo ntago bwemewe.

Iki kibazo kigizwe na parts ebyiri, aho buri subtask ijyanye na **Part I** cyangwa **Part II**. Wemerewe gukora subtasks mu buryo ushatse. Bishatse kuvuga ko udategetswe kurangiza Part I mbere yo gukora Part II.

Part I

Mu gihe ufite description y' urutonde rw' imisozi, shaka umubare wa mythical triples.

Implementation Details

Ugomba gukora implementation y' ibi bikurikira.

```
long long count_triples(std::vector<int> H)
```

- H : array ifite length N , yerekana uburebure bwa peaks.
- Iyi function ihamagarwa kuri buri test case.

Iyi function ingomba kureturninga integer T , umubare wa mythical triples ziri muri mountain range.

Constraints

- $3 \leq N \leq 200\,000$
- $1 \leq H[i] \leq N - 1$ kuri buri i aho $0 \leq i < N$.

Subtasks

Part I ifite amonota 70.

Subtask	Score	Additional Constraints
1	8	$N \leq 100$
2	6	$H[i] \leq 10$ kuri buri i aho $0 \leq i < N$.
3	10	$N \leq 2000$
4	11	Uburebure ntaho bugenda bugabanuka. Bishatse kuvuga, $H[i - 1] \leq H[i]$ kuri buri i aho $1 \leq i < N$.
5	16	$N \leq 50\,000$
6	19	Nta zindi constraints.

Example

Tekereza duhamagaye iyi function

```
count_triples([4, 1, 4, 3, 2, 6, 1])
```

Hari 3 mythical triples muri mountain range:

- Kuri $(i, j, k) = (1, 3, 4)$, uburebure $(1, 3, 2)$ buhwanye na pairwise distances $(2, 3, 1)$.
- Kuri $(i, j, k) = (2, 3, 6)$, uburebure $(4, 3, 1)$ buhwanye na pairwise distances $(1, 4, 3)$.
- Kuri $(i, j, k) = (3, 4, 6)$, uburebure $(3, 2, 1)$ buhwanye na pairwise distances $(1, 3, 2)$.

Bishatse kuvuga ko function ireturninga 3.

Menya ko indices $(0, 2, 4)$ zidakora mythical triple, kubera ko heights $(4, 4, 2)$ zidahuye na pairwise distances $(2, 4, 2)$.

Part II

Ugomba gukora mountain ranges zifite mythical triples nyinshi. Iyi part igizwe na 6 **output-only** subtasks zifite **partial scoring**.

Kuri buri subtask, ufite two positive integers M and K , kandi ugomba gukora mountain range ifite **at most** M peaks. Niba solution yawe ifite **at least** K mythical triples, urabona amanota yuzuye kuri iyi subtask. Bitabaye ibyo, amanota ubona araba ahwanye n' umubare wa mythical triples solution yawe ifite.

Mpenya ko solution yawe ingomba kuba ifite valid mountain range. Muri make, mu gihe solution yawe ifite N peaks (N must satisfy $3 \leq N \leq M$). Then, uburebure bwa peak i ($0 \leq i < N$), bugaragazwa na $H[i]$, bugomba kuba integer iri hagati ya 1 na $N - 1$, inclusive.

Implementation Details

Hari uburyo buburi bwo gutanga solution yawe, kandi wakoresha ubwo ushatse kuri buri subtask.:

- **Output file**
- **Procedure call**

Gusubmittinga ukoresheje **output file**, kora una submittinge text file muri iyi format:

```
N
H[0] H[1] ... H[N-1]
```

Gusubmittinga ukoresheje **procedure call**, ugomba gukora ibi bikurikira.

```
std::vector<int> construct_range(int M, int K)
```

- M : Umubare munini ushoboka wa peaks.
- K : Umubare dushaka wa mythical triples.
- Ibi bikorwa inshuro imwe kuri buri subtask.

Iyi function ireturninga array H ifite length N , yerekana heights za peaks.

Subtasks and Scoring

Part II ifite 30 points. Kuri buri subtask, M na K ntago bihinduka kandi byatanzwe muri table ikurikira:

Subtask	Score	M	K
7	5	20	30
8	5	500	2000
9	5	5000	50 000
10	5	30 000	700 000
11	5	100 000	2 000 000
12	5	200 000	12 000 000

Kuri buri subtask, mu gihe solution yawe idakora valid mountain range, score yawe izaba 0 (ubone na output ivuga ngo: `Output isn't correct` muri CMS).

Bitabaye ibyo, reka T ibe number of mythical triples muri solution yawe. Then, score ya subtask izaba:

$$5 \cdot \min \left(1, \frac{T}{K} \right).$$

Sample Grader

Parts I na II zikoresha same sample grader program, umurongo wambere wa input niwo utadukanya two parts.

Input format for Part I:

```
1
N
H[0] H[1] ... H[N-1]
```

Output format for Part I:

```
T
```

Input format for Part II:

```
2
M K
```

Output format for Part II:

N

H[0] H[1] ... H[N-1]

Menya ko output ya sample grader ihwanye na required format ya output file ya Part II.